Stepwise

Milestone 2 · MLOps Infrastructure

AC215 · Fall 2026 · Live implementation verified 11 October 2026

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1. Scope and milestone coverage

Stepwise helps IELTS Academic Writing Task 2 learners revise argumentative essays. Users type or paste a question and draft, receive four criterion scores with specific improvements, then enter a target band, position and ideas to generate a complete model essay with a separate assessment. Task 2 offers a consistent argumentative task for rubric-based analysis and later adaptation.

MS2 delivers a working Google Gemini application, official-source RAG, a reproducible container pipeline and real execution evidence. The text-only project follows the LLM track; image upload and OCR are outside scope.

Project links

GitHub repository: https://github.com/dwcqwcqw/stepwise

MS2 submission branch: https://github.com/dwcqwcqw/stepwise/tree/milestone2

Live frontend: https://stepwise-one.vercel.app

Requirements: https://harvard-iacs.github.io/2026-AC215/milestone2/

Requirement 1 — Virtual environment (Objective + Deliverable)

Implementation: Cloud Shell provides the Linux VM for Docker development; the deployed API runs in a Cloud Run container. docs/setup.md documents prerequisites, authorization, build, startup and troubleshooting.

Evidence: Figure 1 and docs/evidence/cloud-running.png show the running development environment; pipeline.log records healthy API/web/Qdrant and successful data stages. deploy/README.md explains production setup.

Requirement 2 — End-to-end containers (Objective + Deliverables)

Implementation: services/{ingest,chunk,index,api,web} each has a Dockerfile, pyproject.toml and uv.lock. compose.yaml orders ingestion → chunking → indexing → API → web, with Qdrant health checks. After the documented environment/auth setup, make run starts the complete pipeline.

Data versions: ingest uses the SHA-256-locked official sources (snapshot 3aea32a1828dd022); chunk produces snapshot 52814016b8103614; index builds

2da4e8eda6a331b5; API/web consume that index through the API. Source/chunk/index manifests record lineage and dependency/model versions.

Evidence: docs/setup.md supplies exact commands; docs/evidence/pipeline.log, sample_input.json, feedback_output.json and verification.json demonstrate input → retrieved context → real Gemini output. Deployment uses the same frozen 131-point index.

Requirement 3 — LLM RAG (Objective + Deliverables)

Implementation: services/ingest/main.py collects official PDFs; services/chunk/main.py preserves complete rubric cells and detailed checks; services/index/main.py embeds them into Qdrant. shared/retrieval.py combines mandatory Task 2 rubric coverage with semantic guidance retrieval.

Evidence: docs/evidence/retrieval_example.json contains the query, filters and returned source text/page links; rag_verification.json checks all 40 Task 2 criterion-band cases. Section 3 explains chunking, embeddings, database integration and the complete retrieval example.

Requirement 4 — Vision / other modalities

Not applicable to this text-only LLM project. Users type or paste essays. The 8,049 candidate records and open-model adaptation are reserved for MS3; they are excluded from MS2 retrieval.

Requirement 5 — App mock-up and working skeleton (Objective + Deliverable)

Implementation: docs/wireframe.md and Figures 2–5 cover Draft → Feedback → Goals → Model Essay. services/web and services/api implement real UI/backend interaction. MS1 refinements include removing image/OCR input and decorative panels, expanding the 20 official detail checks, and separating target-band generation from target-blind assessment.

Evidence: the public frontend supports real grading, criterion-specific advice/examples, official citations and full essay generation with a separate assessment. Screenshots, model outputs and public request logs are under docs/evidence.

Submission and TF presentation

All files are supplied on milestone2. Submit its full final commit hash on Canvas by 10:00 PM ET, October 20; this report does not submit to Canvas. docs/slides/Stepwise_MS2_Presentation_Final.pptx supports the 15-minute TF presentation and Q&A. Every member must understand the code, configurations, infrastructure, tests and design choices.

2. Architecture and cloud environment

Data: ingest → chunk → embed/index → Qdrant. Application: browser → web gateway → FastAPI → retrieved official context → Vertex AI Gemini → validated feedback. Successful data stages and health checks gate startup.

Each custom service uses Python 3.12.11 and frozen uv 0.6.4 environments; the Compose Qdrant server uses v1.15.4. Compose API/web bind to loopback and the database has no published host port. The dedicated project stepwise-ac215-20261010 is active under Harvard's self-paid folder, with billing and Vertex AI enabled. Cloud Shell supplies the temporary development VM; the stable public application uses Vercel and a private Cloud Run API.

3. Official knowledge and reproducible RAG

Only three official resources enter MS2. Downloads are SHA-256 checked against data/sources.lock.json. Full PDFs and extracted text stay in ignored runtime storage.

- IELTS Writing Band Descriptors (May 2023): all nine pages, both tasks and bands 0–9; 80 criterion-band cells, including the merged Band 0 rule and limiting features.
- IELTS Writing Key Assessment Criteria: definitions and 20 detailed checks—Task Response 5; Coherence and Cohesion 5; Lexical Resource 6; Grammatical Range and Accuracy 4.
- British Council: Answer the Question: ten pages of official question-analysis and relevance guidance. This older teaching material supplements the current descriptors.

The index has 131 chunks: 80 band cells, 20 detailed checks, five definitions, seven rating rules, nine descriptor pages and ten teaching pages. Whole cells and bullets preserve qualifications; overlap is zero. Page chunks retain full PDF coverage. Metadata records task, criterion, band, page, source URL/hash and stable chunk ID.

Both tasks are indexed for complete source coverage; scoring retrieves only Task 2/shared rules. A mandatory filtered retrieval obtains all 40 Task 2 band cells, all detailed checks and applicable rules. Semantic retrieval adds supplementary teaching context. This prevents top-k search from omitting a criterion or lower band. Generation receives the target-band descriptors; assessment receives the complete band ladder.

FastEmbed 0.7.3 runs BAAI/bge-small-en-v1.5 as CPU ONNX: 384 dimensions and cosine distance. Qdrant stores collection stepwise_2da4e8eda6a331b5. Source snapshot: 3aea32a1828dd022; chunk snapshot: 52814016b8103614; index: 2da4e8eda6a331b5. The manifest records the actual weight hash and dependency versions; stable UUIDs support idempotent upserts.

Retrieval example: POST /api/retrieve with criterion TR, target_band 6.5 and a query about developing examples returns only Task 2 TR descriptors for bands 6 and 7, with text, source/page links and similarity values. The request and complete response are saved in docs/evidence/retrieval_example.json. Similarity selects context; it is not an essay score.

The 8,049 candidate essays are reserved for MS3 fine-tuning. MS2 loads zero candidate records; only a future-use manifest is retained in data/ms3. A regional British Council download block was resolved by caching the exact official PDF, with its locked hash still verified.

4. Real model workflow and application

The verified model is gemini-3.8-flash on Google Vertex AI. The server gives each exact essay span a short evidence ID; the model selects IDs rather than copying text or opaque rubric hashes. The server resolves original passages and attaches the retrieved awarded/next-band descriptors and official detail records. Validation still requires four criteria, all 20 checks, valid rubric IDs, exact excerpts, a rationale, next step and before/after example. One repair attempt receives the prior answer and field-level errors. Diagnostic logs exclude learner text.

The assessor sees only the question, essay and official context. The target band is withheld. Four integer criterion scores are equally weighted and rounded to a half-band Task 2 estimate. Detailed checks are qualitative observations, not separately scored bands.

The four screens are Draft → Feedback → Goals → Model Essay. Generation uses the target, position and requested ideas; a fresh assessment receives only the question and resulting essay. The UI separates target and assessed scores, official citations and idea-coverage notes. Learner text is not persisted by Stepwise; Google receives it for inference. Compose uses short-lived gcloud tokens in a private memory-backed directory. Production uses Vercel OIDC and the Cloud Run runtime identity; it requires no running Cloud Shell token helper.

5. Verification and exact reproduction

Twenty-nine local tests passed after the evidence-reference repair, including preservation of Unicode/spelling errors, bounded exact spans, invalid-reference rejection, repair context and fail-closed behavior. The earlier Compose integration receipt passed 12 real workflow checks; a separate RAG receipt verified 131 points, 80 band cells, 20 detailed checks, all 40 criterion-band retrieval cases and the 6.5 boundary. Historical receipts retain their original implementation version; the production regression evidence is recorded separately.

Observed outputs: the 273-word sample received 8.0; a distinct weak draft received 3.0. The generated essay requested 7.0 and received 9.0 in a separate call. The four-call integration run recorded 22.4 seconds for draft feedback. These are observed outputs, not benchmark accuracy, guaranteed target matching or a latency distribution. Model version, response IDs, token usage and source/index lineage are in docs/evidence.

```
git clone --branch milestone2 https://github.com/dwcqwcqw/stepwise.git
cd stepwise
export GOOGLE_CLOUD_PROJECT=stepwise-ac215-20261010
export STEPWISE_AUTH_DIR=/dev/shm/stepwise-auth
python3 scripts/cloud-auth.py > /tmp/stepwise-auth.log 2>&1 &
make run
```

Prerequisites: Docker/Compose, make, Python 3, an authorized gcloud session and an enabled billing/Vertex project. Open localhost:8080 or Cloud Shell Web Preview on port 8080. docs/setup.md contains full configuration and troubleshooting. make verify repeats real model

calls; make evidence captures existing outputs/logs; make test runs without Google calls. The Cloud Shell preview is temporary.

6. Public deployment and verification

Live application: https://stepwise-one.vercel.app is public without login. Bei Li's complete implementation is on the default bei-li branch; milestone2 retains the existing course delivery snapshot. main and the other named member branches have empty starting contents for separate team development. The frontend uses source 95b921c8e19d81280bb7c15d173cdf19c4f841c8; the repaired private API uses b97467fa4dddf9a55e0a7408a5957605f540e766, deployed as stepwise-api-00002-tnt with 100% traffic. deploy/README.md contains production instructions.

Cloud connection: Vercel → short-lived OIDC identity → private Cloud Run → Vertex AI Gemini. The API loads the frozen 131-point official rubric index and pinned embedding model. Dedicated runtime, invoker and build identities separate permissions. No Google credentials are embedded in frontend code; Cloud Shell can stop without stopping the deployed service.

Production regression on 11 October 2026: a 271-word authored draft with grammar errors, curly punctuation and doubled spaces returned TR 8, CC 8, LR 8 and GRA 7 (overall 8.0). All 20 detailed-check quotes and four main quotes matched the original. Generation then produced 299 words for target 7.0; a separate assessment returned TR 8, CC 9, LR 9 and GRA 9 (overall 9.0), with all 20 detailed quotes verified. Both requests returned HTTP 200; recorded backend latencies were 16.6 and 14.9 seconds respectively. Twenty-nine local tests passed. See docs/evidence/reference-regression-*. Earlier production and Compose receipts remain versioned separately. Target overshoot still requires calibration. Vercel Git auto-deployment is not configured; frontend files were uploaded through its API.

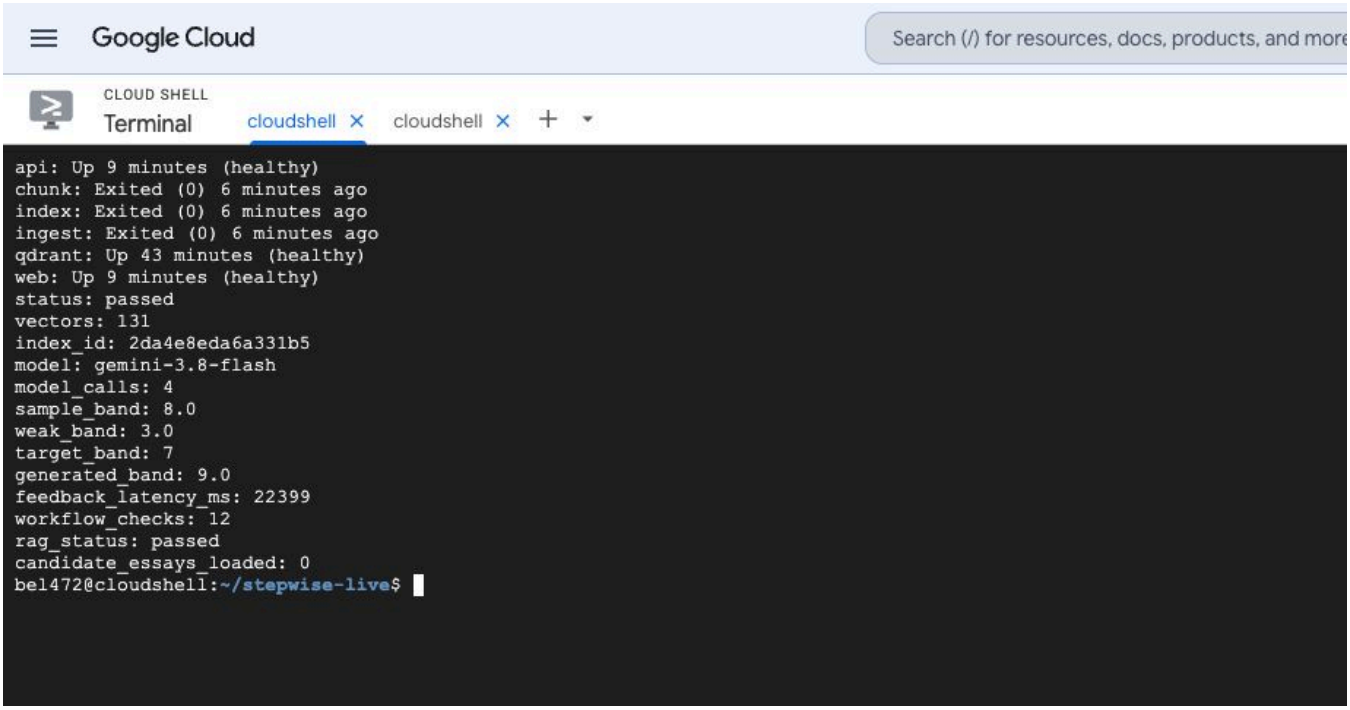
7. MS3 development and references

Next: audit reliable Task 2 adaptation data, label provenance and prompt-family leakage; compare prompting, official RAG and an open-model LoRA baseline. Track criterion MAE, within-0.5-band agreement, repeated-run variance, target-band deviation, latency and token cost. Current samples establish integration; high generated scores and target overshoot require held-out calibration. Open-model deployment, request limits and load evaluation follow in later milestones.

Reference implementations: cheese-app-v0 informed pipeline separation; babykev informed reproducible environments and explicit checks. Stepwise supplies its own IELTS parsing, retrieval contracts and UI. Team members should understand the entire implementation.

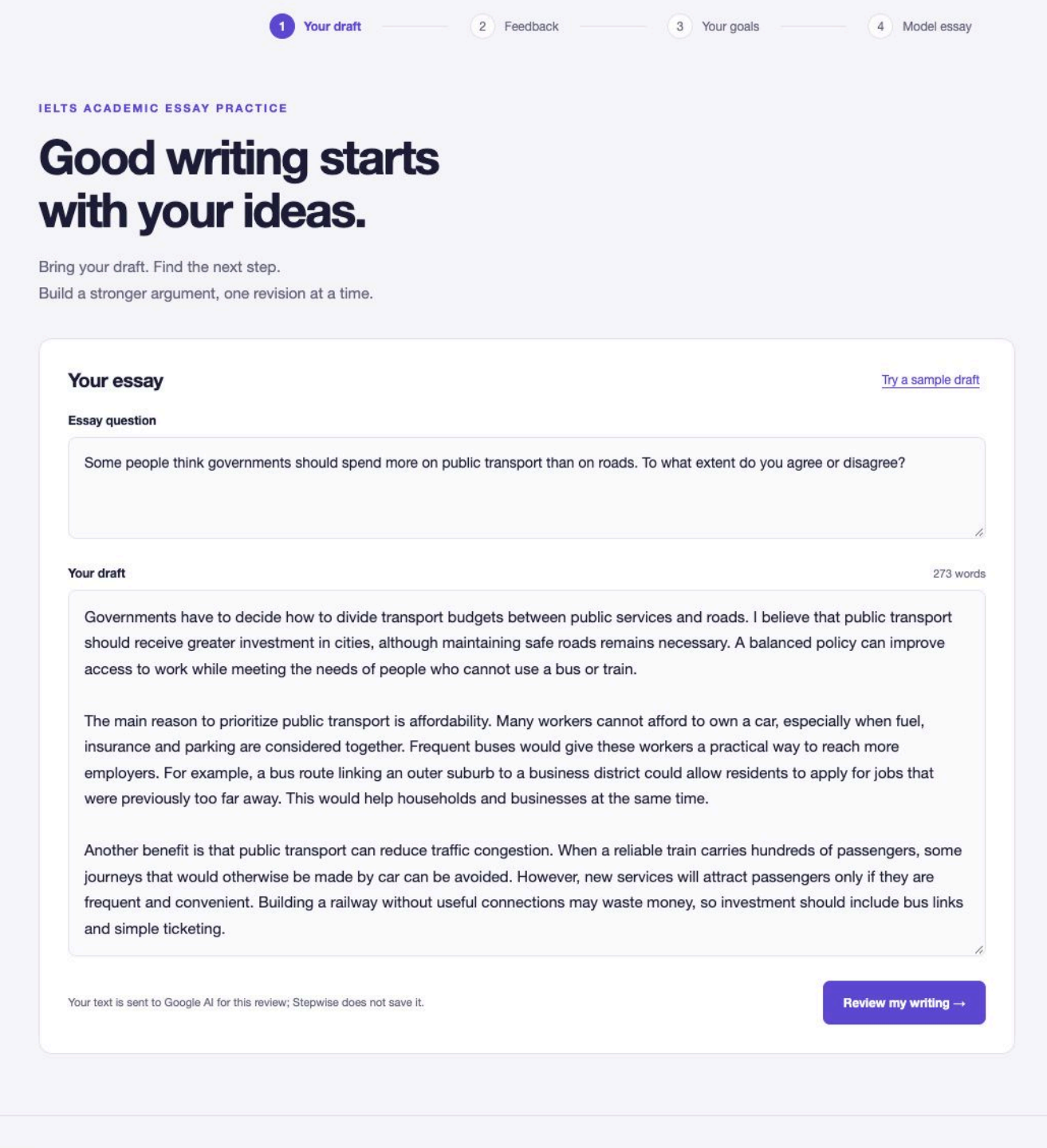
Code, receipts and presentation: https://github.com/dwcqwcqw/stepwise/tree/milestone2. Earlier Compose verification: c4522d75b45c2ae32a73b0b5c133ccfcb5e43a71. Production evidence repair: b97467fa4dddf9a55e0a7408a5957605f540e766. The final branch also includes this report and refreshed evidence; obtain its final Canvas hash with git rev-parse HEAD.

Figure 1. Running cloud environment



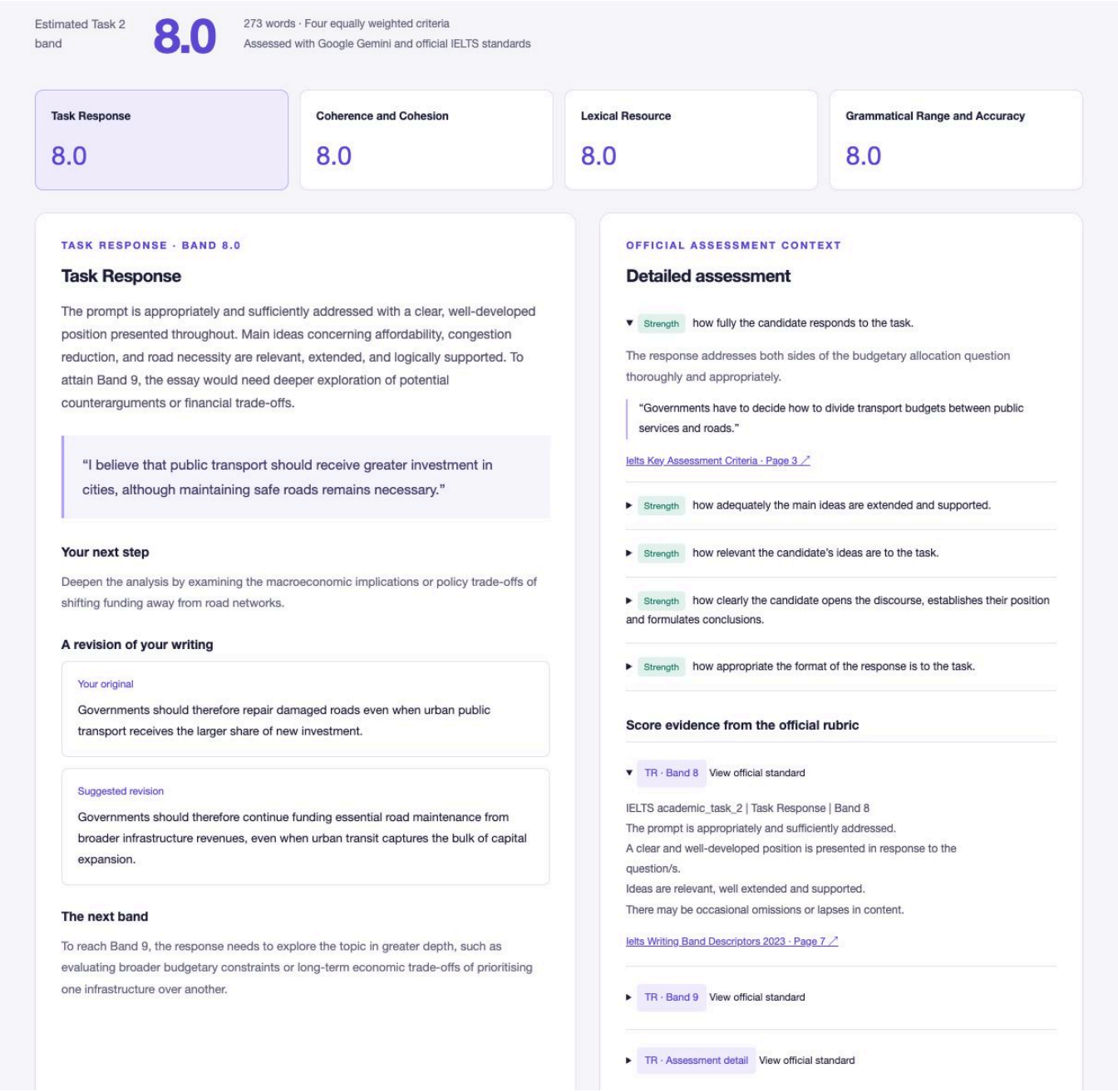
Actual Cloud Shell VM: API, web and Qdrant are healthy; ingest/chunk/index exit 0. The real model integration passes with 131 vectors. Full logs and manifests accompany the repository.

Figure 2. Your draft



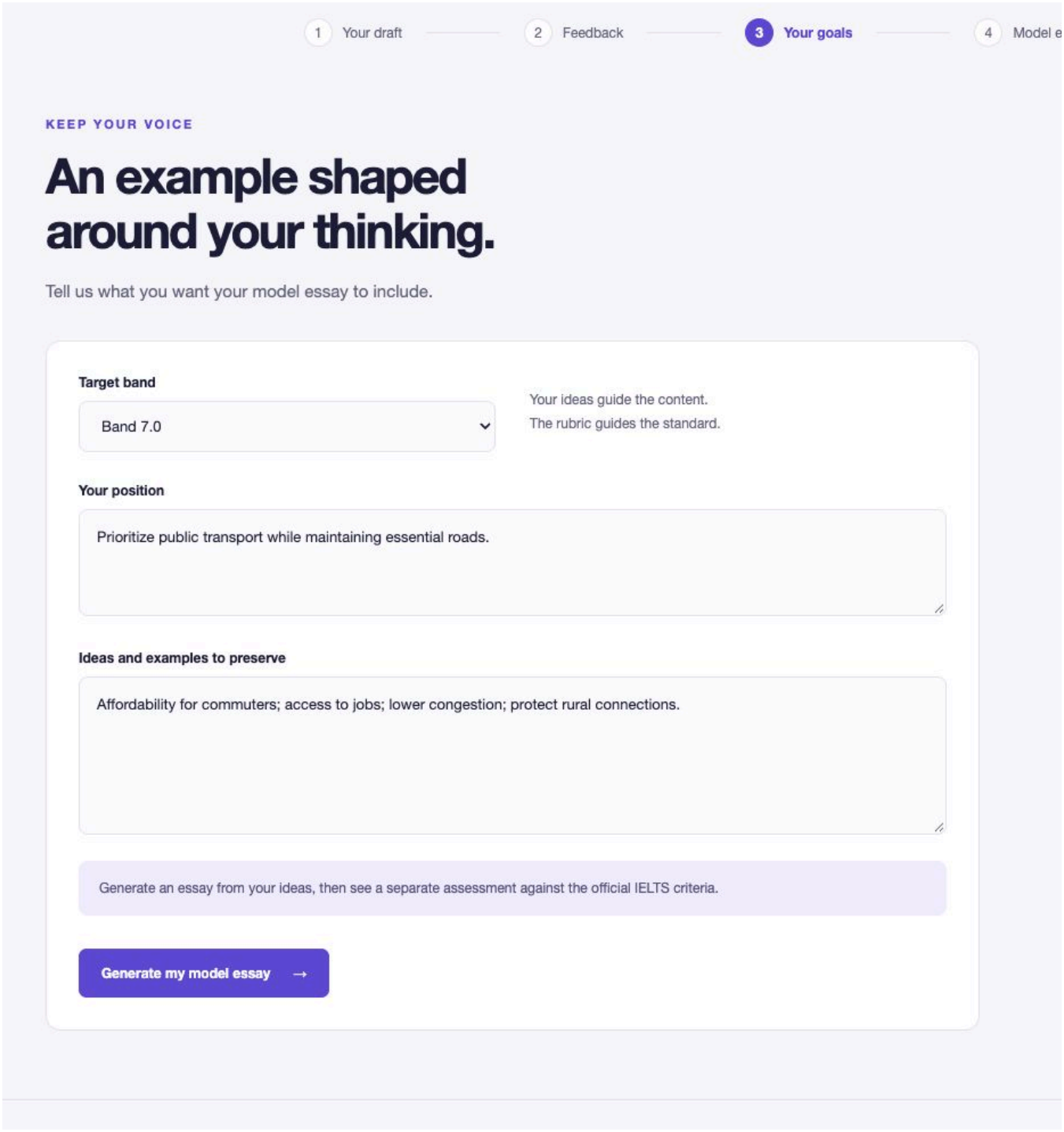
One editor accepts a typed or pasted question and essay, with a sample draft and live word count. The decorative panel, initial target sidebar and image/OCR controls are absent.

Figure 3. Criterion feedback and official detail



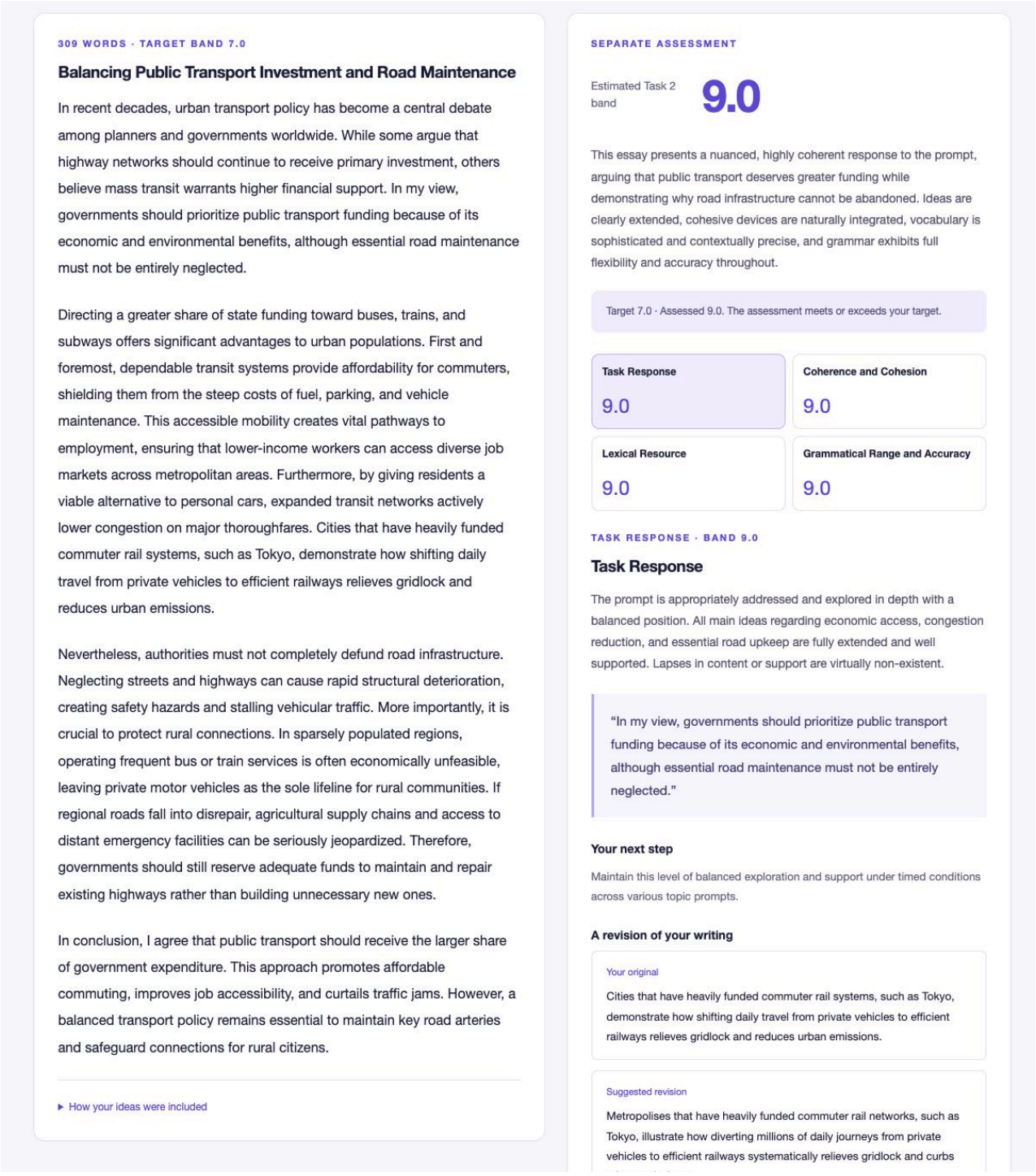
The live sample receives 8.0. Exact essay evidence, a specific revision and an example appear beside official detailed checks and retrieved band descriptors. Source links open the original PDF page.

Figure 4. Your goals



The learner specifies a target band, position and ideas to preserve before generating the model essay.

Figure 5. Generated essay and separate assessment



309 WORDS · TARGET BAND 7.0

Balancing Public Transport Investment and Road Maintenance

In recent decades, urban transport policy has become a central debate among planners and governments worldwide. While some argue that highway networks should continue to receive primary investment, others believe mass transit warrants higher financial support. In my view, governments should prioritize public transport funding because of its economic and environmental benefits, although essential road maintenance must not be entirely neglected.

Directing a greater share of state funding toward buses, trains, and subways offers significant advantages to urban populations. First and foremost, dependable transit systems provide affordability for commuters, shielding them from the steep costs of fuel, parking, and vehicle maintenance. This accessible mobility creates vital pathways to employment, ensuring that lower-income workers can access diverse job markets across metropolitan areas. Furthermore, by giving residents a viable alternative to personal cars, expanded transit networks actively lower congestion on major thoroughfares. Cities that have heavily funded commuter rail systems, such as Tokyo, demonstrate how shifting daily travel from private vehicles to efficient railways relieves gridlock and reduces urban emissions.

Nevertheless, authorities must not completely defund road infrastructure. Neglecting streets and highways can cause rapid structural deterioration, creating safety hazards and stalling vehicular traffic. More importantly, it is crucial to protect rural connections. In sparsely populated regions, operating frequent bus or train services is often economically unfeasible, leaving private motor vehicles as the sole lifeline for rural communities. If regional roads fall into disrepair, agricultural supply chains and access to distant emergency facilities can be seriously jeopardized. Therefore, governments should still reserve adequate funds to maintain and repair existing highways rather than building unnecessary new ones.

In conclusion, I agree that public transport should receive the larger share of government expenditure. This approach promotes affordable commuting, improves job accessibility, and curtails traffic jams. However, a balanced transport policy remains essential to maintain key road arteries and safeguard connections for rural citizens.

▸ How your ideas were included

SEPARATE ASSESSMENT

Estimated Task 2 band **9.0**

This essay presents a nuanced, highly coherent response to the prompt, arguing that public transport deserves greater funding while demonstrating why road infrastructure cannot be abandoned. Ideas are clearly extended, cohesive devices are naturally integrated, vocabulary is sophisticated and contextually precise, and grammar exhibits full flexibility and accuracy throughout.

Target 7.0 · Assessed 9.0. The assessment meets or exceeds your target.

Task Response	Coherence and Cohesion
9.0	9.0
Lexical Resource	**Grammatical Range and Accuracy**
9.0	9.0

TASK RESPONSE · BAND 9.0

Task Response

The prompt is appropriately addressed and explored in depth with a balanced position. All main ideas regarding economic access, congestion reduction, and essential road upkeep are fully extended and well supported. Lapses in content or support are virtually non-existent.

> "In my view, governments should prioritize public transport funding because of its economic and environmental benefits, although essential road maintenance must not be entirely neglected."

Your next step

Maintain this level of balanced exploration and support under timed conditions across various topic prompts.

A revision of your writing

Your original

Cities that have heavily funded commuter rail systems, such as Tokyo, demonstrate how shifting daily travel from private vehicles to efficient railways relieves gridlock and reduces urban emissions.

Suggested revision

Metropolises that have heavily funded commuter rail networks, such as Tokyo, illustrate how diverting millions of daily journeys from private vehicles to efficient railways systematically relieves gridlock and curbs

Detail from the running model-essay screen: complete prose and a separate four-criterion assessment. Target 7.0 and assessed 9.0 remain distinct. The full screen and JSON output include detailed checks, revisions and linked standards.